# AMIT PAUL

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GitHub LeetCode LinkedIn Bitbucket Codeforces

**Know more about me** → https://amitpaul-portfolio.netlify.app/

## PROFESSIONAL EXPERIENCE

**Idlewild Digital | Backend Developer (Remote)** Jan 2024 – Present

- **System Development & Efficiency:** Engineered "Crabarian," a custom data ingestion engine using Python, reducing manual operational load by 35+ hours/week through automated ETL pipelines.
- **Web Application Development:** Built a modular Learning Management System (LMS) serving 5,000+ users, implementing Role-Based Access Control (RBAC) and secure data handling protocols.
- **Performance Optimization:** Debugged and resolved system latency issues in PostgreSQL. Improved query efficiency by 40% by eliminating N+1 bottlenecks using Django's `select_related` and `prefetch_related`.
- **Full Stack Implementation:** Developed "American Provost Connect," a community platform. Integrated Vue.js (Frontend) with Django (Backend) to enable interactive event management and forum features.
- **Code Health & Best Practices:** Enforced code quality and testability by implementing unit tests and conducting weekly code reviews to ensure architectural consistency across the team.

**Idlewild Digital | Python Intern** Aug 2023 – Dec 2023

- **Product Lifecycle:** Owned the end-to-end SDLC of the "Daily Updates" internal reporting system, deploying the solution in a Linux/Unix environment.
- **Rapid Delivery:** Promoted to Backend Developer in 5 months for demonstrating strong grasp of software engineering principles and modular design patterns.

## TECHNICAL SKILLS

**Languages:** Python (Advanced), JavaScript (ES6+), Java, C++, C, SQL.
**Full Stack & Web:** Django, Django REST Framework, Vue.js, HTML5, CSS3, REST APIs.
**Infrastructure & Tools:** Linux/Unix Environments, Docker, Git, CI/CD, PostgreSQL.
**Core Competencies:** Data Structures & Algorithms, System Design, Performance Optimization.

## INTERNSHIPS AND ATTACHMENTS

**Grameenphone | Data Science & Engineering Trainee**(Industrial Attachment) Jan 2025 – Feb 2025

- **AI & Information Retrieval:** Developed a Retrieval-Augmented Generation (RAG) chatbot API. Utilized vector embeddings and Python to automate customer support queries with high accuracy.

**University Certificate Automation System** | Full Stack Lead

## PROJECTS & RESEARCH

**University Certificate Automation System** | Full Stack Lead

- Designed, built, and pitched a full-stack certificate platform (designed to scale to 10,000+ students) to the Khulna University ICT Cell, outlining its architecture, security, and deployment plan.
- **Links:** Backend | Frontend | Demo | Presentation

**Bearing Fault Diagnosis Research** | Thesis Researcher (Ongoing)

- Conducting comparative analysis of Vision Transformers (ViT) and Time Series Transformers for signal processing, optimizing for noise robustness and cross-dataset accuracy.

## EDUCATION

**Khulna University** Graduated: January 2026
B.Sc. Engg. in Electronics and Communication Engineering

- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Computer Networks, OOP.
- **Awards: ICPC Asia Dhaka 2021** (Competitive Programming), Active Solver on Codeforces/LeetCode.